	W3	W12	W16
Original	,the numbers glaring back at me like an unsolvable riddle	glaring back at me like an unsolvable riddle	,the numbers glaring back at me like an unsolvable riddle
Category	Cliche	Cliche	Unnecessary/Redundant Exposition
Edited	. The numbers stared back	barreling over one another as they raced to some unseemly height.	-
Original	and an unsettling sense of mystery that gnawed at me more than the inexplicable weight itself	Her words felt like a placeholder for an answer neither of us had yet. I walked out with a slip for blood tests and an unsettling sense of mystery that gnawed at me more than the inexplicable weight itself.	-
Category	Purple Prose	Cliche	-
Edited	-	But when I saw her turn to go, whispering in the halls with a colleague, I knew there was still something she had yet to tell me.	-

Table 7: Original spans selected by 3 writers from the same paragraph. '-' denotes the span was deleted while editing

We analyze the editing process by examining *edit operations*: insertion, deletion, or replacement. An edit is an insertion if it deletes no characters or adds 40+ characters net. Conversely, it's a deletion if it adds no characters or removes 40+ characters net. All other edits are tagged as replacements. We choose a threshold of 40 characters (roughly 10 words), to avoid labeling edits with minor length changes ("a" to "the") as insertions or deletions. Figure 4a shows edit operations by participant for each paragraph. Replacements are most frequent (74%), followed by deletions (18%) and insertions (8%). Editing styles vary: some participants primarily use replacements (W2, W9, W10, W16, W18), while others employ deletions more often (W1, W5, W7, W8, W13, W17). Insertions are uncommon across all participants. To quantify meaning-preserving vs. meaning-changing edits, we calculate semantic similarity between original and edited text using BERT score[118]. Using a threshold of 0.6 [7], we classify edits with similarity > 0.6 as meaning-preserving. Of 6468 non-deletion edits, 70% are meaning-preserving, with the rest meaning-changing. This finding supports our Design Principle 2. Figure 5c shows the distribution of the semantic similarity scores for the edits in LAMP. [8]. While the distribution is bimodal (with peaks around 0.6 and 0.75), these modes don't align well with the 0.6 threshold. This suggests that meaning preservation versus meaning change exists more on a continuum rather than as two clearly distinct categories.

The annotation interface allowed participants to provide Initial and Final Writing Quality Scores (IWQS and FWQS) for each paragraph, ranging from 1 to 10. Figure 4b shows the distribution of these scores for each participant, revealing significant variability (e.g., W1's median IWQS is 7, W18's is 2). Calibration of writing quality scores is a known challenge, and we follow prior work in normalizing the scores into *z-Scores* by subtracting the mean, dividing by the standard deviation of the scores for each participant [45, 73], and re-scaling them to the 1 to 10 range. Subsequent analyses use these normalized scores.

[7]This threshold was decided by manually analyzing 100 edits

[8]However we note that with a threshold of 0.6, this appears to be quite a fuzzy/arbitrary distinction rather than a clear separation

We compute an *edit distance* between the original LLM-generated text and the final edited text, by calculating a character-level Levenshtein distance [66] between the two strings of texts. The edit distance measures the "amount of editing work" performed by a writer. Figure 4d shows a negative correlation between edit distance and IWQS (Pearson's $r = -0.31$), indicating that higher perceived text quality (high IWQS) requires less editing, while lower IWQS necessitates more editing.

Figure 4c shows the average IWQS for each LLM on creative non-fiction and fiction writing tasks. Writers were unaware of which model generated each text, and tasks were shuffled to avoid bias. This analysis estimates the writing quality of the three models in both domains. Comparing model scores, we find no significant difference in writing quality across the three models. GPT-4o and Claude 3.5-Sonnet perform slightly better on creative non-fiction instructions (average 5.2) compared to Llama3.1-70B (5.0), though the difference is not statistically significant. All models show a slight decrease in performance for fictional instructions, with an average IWQS of 4.5. This suggests fiction writing may be more challenging for LLMs than creative non-fiction. These findings differ from task-oriented benchmarks that reveal performance gaps between models in areas like factual or logical reasoning. **Our results indicate that, when it comes to creative writing, writers perceived no significant qualitative differences among the texts generated by large language models (LLMs) such as GPT-4, Claude 3.5 Sonnet, and Llama 3.1 70B.**

Figure 5a displays edit categories applied by writers to texts from three LLMs. The distribution is similar across models, with the most common categories being *Awkward Word Choice and Phrasing* (28%), *Poor Sentence Structure* (20%), *Unnecessary/Redundant Exposition* (18%), and *Clichés* (17%). Minor differences include GPT-4o using more purple prose and Llama3.1-70B generating more unnecessary exposition. Overall, **LLMs across the three model families exhibit similar idiosyncrasies that are edited out in similar proportions by professional writers.** Figure 5b illustrates the relationship between edit categories and IWQS. Higher IWQS scores correspond to fewer total edits, with texts rated 2 averaging

10.2 edits and those rated 10 receiving 2.4 edits, confirming that higher-quality texts need less editing. This trend however varies across edit categories: "Unnecessary/Redundant Exposition" and "Lack of Specificity and Detail" remain relatively constant, while the number of "Awkward Word Choice and Phrasing" and "Cliché" edits decrease as IWQS increases, suggesting a stronger correlation with perceived writing quality.

5.2 *Writers differ greatly in the amount of editing they do*: But to what extent?

The writer's approaches to editing vary based on personal or organizational philosophy. Some prioritize preserving the original voice and make minimal changes to preserve authenticity [109]. Others may take a more interventionist stance, heavily revising to align with their vision or house style. Additionally, some writers might make fewer but more impactful changes, while some might make numerous small revisions. To quantify this, we asked 3 writers (W3, W12, and W16) to edit a subset of the same 50 paragraphs from the LAMP Corpus. As expected, these three writers differed in the amount of editing they did. W3 did 9.4 edits on average while W12 and W16 did 6.0 and 6.3 edits on average. On average the span level precision (see Section 6.1 for more detail on the metric) between the 3 writers was 0.57 suggesting a moderately significant agreement.

Table 7 shows how sometimes writers select the exact same problematic span but assign different categories. A span that both W3 and W16 selected "*, the numbers glaring back at me like an unsolvable riddle*" (Table 7 Row 1) was categorized differently (Table 7 Row 2). Yet both categorizations can be correct interpretations. When one relies on overused phrases or clichés, they often state the obvious or provide information that readers can easily infer implicitly. This results in redundant or superfluous exposition that doesn't add value to the narrative. Other times writers may select the same category but with only partial overlap on the selected span (Table 7 Row 1 W3 vs W12). Looking at (Table 7 Row 4 and 5; W3 vs W12) there is a partial overlap in the selected span "*and an unsettling sense of mystery that gnawed at me more than the inexplicable weight itself*". However, the selected categories are Purple Prose and Cliché respectively. Here again, it should be noted that Purple Prose is a style of writing that can be original or cliché, depending on its usage, context, and frequency. Not all elaborate writing is overused, but when certain ornate phrases or styles become too common, they can cross the line into cliché territory. W16 however did not edit this span.

We also highlight that diversity in edits among writers such as selecting different spans or rewriting it in an individualistic style is a positive aspect that prevents homogenization while still improving LLM-generated text as shown by our results in Section 6.2.

5.3 Are there any specific stylistic idiosyncrasies in LLM generated responses?

Recent work from Shaib et al. [103] uses syntactic patterns with Part-of-speech [9] as abstract representations of texts, that can capture more subtle repetitions than mere text memorization. They find that language models tend to use repetitive syntactic templates more often than humans and these patterns can help evaluate style memorization in language models. Following their experiments we consider Part-of-speech templates of length $n \epsilon \{5, 6, 7, 8\}$ in LLM-generated responses as well as the original seed human-written paragraphs (Table 1). We looked at the 50 most common templates in LLM-generated responses and found that 15 templates do not occur as frequently in original human-written seed paragraphs. Table 8 shows representative sequences corresponding to particular syntactic patterns present in higher proportion in LLM-generated responses. These sequences constitute categories of **Clichés, Unnecessary/Redundant Exposition or Poor Sentence Structure** and are often heavily edited by writers in our study.

To better understand idiosyncrasies, we examined awkward words/phrases occurring disproportionately in LLM-generated responses. For instance, Figure 6 shows how a word like *unspoken* occurs in about 15% of LLM-generated responses. Similarly phrases such as *weight of*, *sense of*, *mix of* occur very rarely or not at all in original seed paragraphs (Table 1) while they occur frequently in LLM-generated responses. We also found peculiar and uncommon phrases generated by LLMs across several responses such as *air was thick*, *hung in the air*, *eyes darting*, *a sense of unease (grew/growing/settles) in the pit of (her/my) stomach*. The most surprising finding is that all 3 LLMs generate these idiosyncratic words/phrases **suggesting possible overlap/mixture in instruction tuning data across model families or one model trained on synthetic data generated from another model** [120].

6 AUTOMATIC DETECTION AND REWRITING OF LLM IDIOSYNCRASIES

While several automated editing approaches exist for improving LLM outputs at scale, we evaluate their effectiveness specifically for addressing idiosyncrasies in creative writing. Building on Hayes et al. [47] and Scardamalia [102], we develop techniques to separate detection and rewriting tasks, evaluating them using LAMP Corpus annotations. Given automated evaluation limitations for text editing[27], we conduct a large-scale preference annotation study with LAMP Corpus writers, comparing human and LLM-produced edits. To accommodate methods that require training samples, we split our data: 146 of 1057 LAMP Corpus paragraphs for training, the rest for testing.

6.1 Automatic detection of problematic spans in LLM-generated text

We formulate the problem of detecting problematic spans in LLM-generated text as a *multi-span categorical extraction* problem. In other words, given a paragraph of LLM-generated text, the method must output a list of non-overlapping spans present in the original text, and assign a category to each extracted span (from the list of categories of the LAMP Corpus).

To evaluate various methods, we use the span-level precision metric, a common metric used in NLP tasks requiring comparison of extracted spans [96]. Span-level precision measures the degree of overlap between predicted spans and reference or ground truth spans (in our case spans collected as a part of the editing process from writers). The overlap is measured at the character level, such

[9] https://www.sketchengine.eu/blog/pos-tags/

Syntactic Pattern	% of Times Edited	Representative Sequence
DT NN IN NN CC	54%	a mix of pride and, a mix of fear and, a sense of protection and, a sense of wonder and, a means of connection and, a pang of nostalgia and, a pang of disappointment, but, a flicker of hope or, a blend of relaxation and, a blend of curiosity and, the power of storytelling, the power of empathy, and, a web of belonging and [....]
NN IN NN CC NN	35	scene of chaos and destruction, mix of desperation and resolve, mix of relief and gratitude, torn between curiosity and caution, breakfast of bread and jam, sense of calm and normalcy, perception of loyalty and identity, glimmer of fear and vulnerability, meaning of protection and care, story of struggle and resilience, blend of fear and hope, [....]
DT JJ NN IN PRP$	40	a constant reminder of his, the mundane routine of our, the intricate tapestry of its, the subtle shift in their, the potential weight of its, a quiet sigh as her, a small acknowledgment of their, the upcoming chapter of her, a silent battle between his, a complex blend of their, the subtle shift in her, the unspoken plea in her [...]
DT NN IN JJ NN	27	the fabric of daily life, a moment of genuine connection, a life of absolute relaxation, the face of inevitable loss, the weight of past grievances, a state of constant unease, a residue of weary resilience, a sea of unspoken expectations, a mask of controlled concern, the weight of unresolved history, a foundation of silent understanding, [....]
IN DT NN IN NN CC	45	with a mix of wariness and, by the hum of traffic and, in a flurry of pursuit and, into a world of precision and, in a gesture of comfort and, in a storm of pain and, for the sake of stability and, in the rhythm of routine or, in the magic of family and, like the depth of understanding and, with a sense of nuance and [...]

Table 8: Idiosyncratic Sequences following certain syntactic patterns in LLM generated responses that are edited by writers. These syntactic patterns do not occur in the human written seed paragraphs

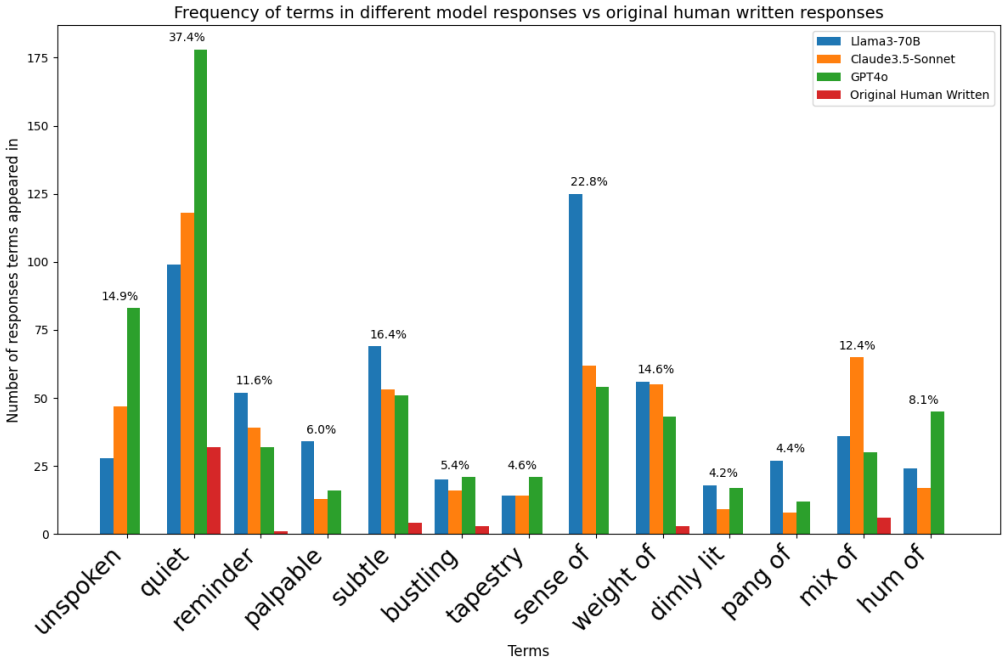

Figure 6: Distribution of peculiar and odd words and phrases occurring in LLM-generated text vs. human-written text in the LAMP Corpus.

that spans that partially overlap will get precision scores that reflect the amount of overlap between the two spans. High span-level precision indicates that the model is precise in identifying the correct boundaries of relevant text spans without over-predicting. We implement two precision metric variants: General and Categorical Precision. General Precision credits span selection regardless of category assignment, while Categorical Precision requires correct category assignment. We use a precision-based metric (like BLEU